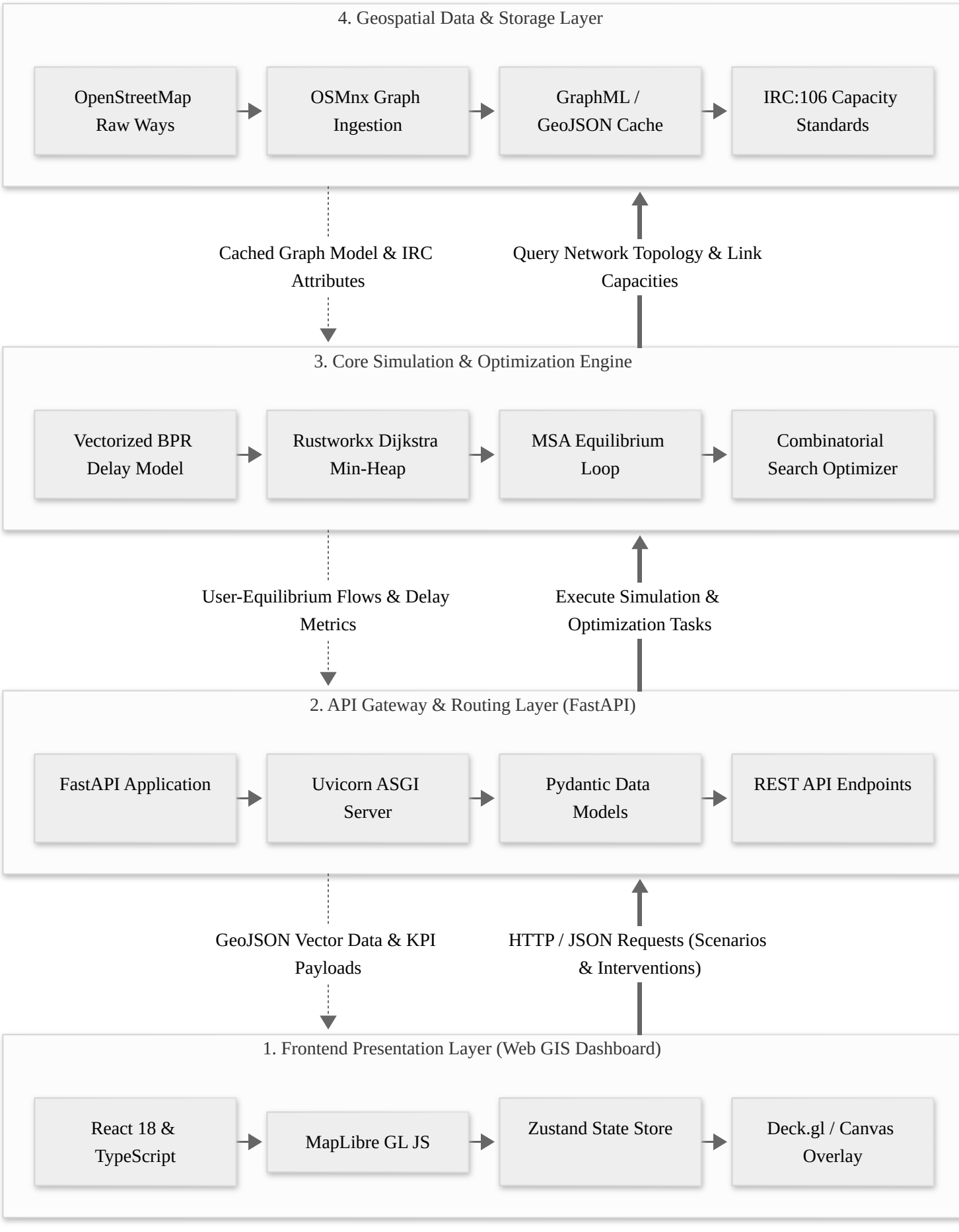

4. Geospatial Data & Storage Layer
OpenStreetMap Raw Ways
OSMnx Graph Ingestion
GraphML / GeoJSON Cache
IRC:106 Capacity Standards
Cached Graph Model & IRC Attributes
Query Network Topology & Link Capacities
3. Core Simulation & Optimization Engine
Vectorized BPR Delay Model
Rustworkx Dijkstra Min-Heap
MSA Equilibrium Loop
Combinatorial Search Optimizer
User-Equilibrium Flows & Delay Metrics
Execute Simulation & Optimization Tasks
2. API Gateway & Routing Layer (FastAPI)
FastAPI Application
Uvicorn ASGI Server
Pydantic Data Models
REST API Endpoints
GeoJSON Vector Data & KPI Payloads
HTTP / JSON Requests (Scenarios & Interventions)
1. Frontend Presentation Layer (Web GIS Dashboard)
React 18 & TypeScript
MapLibre GL JS
Zustand State Store
Deck.gl / Canvas Overlay